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Two piece, T-shaped dispensing tap and method of making same

Abstract

A two piece, molded tap is contemplated. The tap is includes a T-shaped body with a rotatable cap serving as a valve element. Camming means are provided on opposing but concealed faces of the cap to enable the smooth and easy rotation of the cap between open and closed positions while rotating the cap less than one full revolution in comparison to the body which remains fixed to a flexible pouch or container.

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USPC: 222/509

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1339665	12/1919	Otto	251/210	F16K 3/246
2217835	12/1939	Corbin, Jr.	137/630.22	F16K 3/246
3062496	12/1961	Stehlin	251/324	F16K 3/28
3990677	12/1975	Grenier	251/189	F16K 3/246
4619377	12/1985	Roos	222/129	F16K 3/246
6321948	12/2000	Bellon	N/A	N/A
7681764	12/2009	Nini	N/A	N/A
7721755	12/2009	Smith	N/A	N/A
8336743	12/2011	Bellmore	N/A	N/A
8387837	12/2012	Bellmore	222/571	B67D 3/0058
10696536	12/2019	Nini	N/A	N/A
10792631	12/2019	Roberts	N/A	B01F 25/31242
2004/0135113	12/2003	Roos	251/265	B67D 3/045
2005/0104021	12/2004	Meyers	251/81	F16K 35/04
2005/0211726	12/2004	Pritchard	222/105	B67D 3/045
2006/0162785	12/2005	Smith	137/384	B65D 77/067
2022/0041426	12/2021	Nini	N/A	B67D 3/043
2022/0089428	12/2021	Edwards	N/A	F16K 3/246

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
0115164	12/1983	EP	N/A
0272906	12/1987	EP	N/A
1381391	12/1974	GB	N/A
2008207867	12/2007	JP	N/A
6423992	12/2017	JP	N/A

OTHER PUBLICATIONS

Patent Cooperation Treaty (PCT), International Search Report and Written Opinion for Application PCT/US2021/062129 filed Dec. 7, 2021 mailed Feb. 25, 2022, International Searching Authority, US. cited by applicant
European Extended Search Report dated Oct. 9, 2024; European Patent Application No. 21904217.3. 8 pages. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS AND TECHNICAL FIELD (1) This application claims priority to U.S. provisional patent application Ser. No. 63/122,182 filed on Dec. 7, 2020, and is a national stage filing of international patent application PCT/US2021/062129 filed on Dec. 7, 2021 and claiming priority to the aforementioned provisional application. Both of applications are incorporated by reference. (2) This application relates generally to dispensing apparatus and, more specifically, to a two-piece tap assembly for selectively dispensing liquids through a vertical/upright tubular assembly that is attachable to a liquid filled pouch or container.

BACKGROUND
(1) Dispensing taps are often used in combination with pouches, bags, and other fungible, replaceable, and easy-to-fill container systems. “Bag-in-box” and pouch-style liquid containers are particularly useful in the food industry because these relatively pliant and shapeable vessels can be easily placed and replaced in sturdier, prism- or cylindrically shaped containers.
(2) The taps typically have a flange or flat member facing and/or coupled to the inner facing of the bag. In one arrangement that is used preferentially in the food service industry, a lateral/horizontally oriented flow tube connects to a cylindrical dispensing tube with valve controlling means positioned atop the dispensing tube.
(3) U.S. Pat. No. 6,321,948 provides one example of such a tap. A twistable handle opens and closes a valve structure contained in an upright body. U.S. Pat. No. 7,721,755 shows a similar style tap, but with a pair of spiral grooves formed in a knob atop the upright body and camming projections protruding through the grooves. U.S. Pat. No. 8,336,743, as well as Japanese patent publication JP6423992B2 and European patent publication EP0272906A2, all disclose cap and upright body arrangements in which the camming interfaces are concealed beneath the tap. Notably, camming interfaces are preferred over conventional screw threads because the cam typically presents at a steeper pitch that requires less than one full revolution/turn of the cap to open or close the valve.
(4) A different style of dispensing taps can be found in U.S. Pat. Nos. 4,619,377 and 7,681,764. Here, the flow tube extends orthogonally (i.e., horizontally) away the attachment flange so that the valve cap and valve body are all aligned along a common axis. Or stated more simplistically, whereas the previous tap style presents a T-shape, this different style is aligned in a straight line to impart an I-shape appearance. Notably, the sealing faces in an I-style shape are reduced and simplified, although persons using the tap may find the T-shape tap easier to actuate/use, especially to the extent the dispensing port in a T-shape tap is immediately identifiable (i.e., at the bottom most edge of the T-shape), whereas the location of the dispensing port is not as apparent.
(5) Still other styles of dispensing taps having a T-shape are contemplated by international patent application WO2020/053904 and U.S. Pat. No. 10,696,536. These disclosures all show multiple

elements comprising the valve cap and/or camming mechanism. This multiplicity of parts creates complexity and increases the possible modes of failure.

(6) For the sake of clarity and to further highlight and contrast with certain aspects of the invention disclosed herein, all of the aforementioned disclosures are incorporated by reference into this Background section.

(7) In view of the foregoing, a two-piece dispensing tap having a familiar hollow, T-shape and an easy to actuate valve integrally formed as a cap or knob would be welcome. Further, a tap is needed where the camming mechanism is concealed to minimize, if not entirely prevent, debris from accumulating within valve itself. Lastly, an ideal tap has a simple, two piece construction with a stem having a continuous, solid surface and presenting a more pleasing aesthetic that can be actuated by one full turn or less relative to the body.

SUMMARY OF INVENTION

(8) A two piece dispensing tap, and a method of easily injection molding the same, is contemplated. The tap consists solely of a hollow tubular, T-shaped body defining a flow channel and a cap or knob (also referred to as a stem). The cap includes an outer-most skirt and a coaxial cylindrical sealing member positioned therein.

(9) Camming protrusions on the inner and outer facings of one end of the body fit within cooperating camming grooves formed on both of the inner facing of the skirt and an outer facing of the sealing member, both of which extend down from a top closing/sealing panel formed on the cap. The skirt-facing, spherical protrusions are larger (i.e., extend radially away from the sidewall more) in comparison to the rounded, bump-like protrusions interfacing with the sealing member.

(10) Both sets of protrusions conform to corresponding grooves formed on the cap skirt and sealing member, while seats or stopper indentations may positioned within the grooves so as to provide a tactile sensation for the open position. The pitch and positioning of the grooves on the cap also ensure the opening and closing of the valve can be achieved in less than one full turn (and as little as about one half turn) of the cap.

(11) Lastly, the cap's outer diameter largely conforms to the upright member of the body, thereby insuring a sufficiently small footprint (relative to the diameter of the flange on the attachment end of the body) so that the body and tap are easily welded/coupled to a pouch or bag. The outer facing of the sealing cylinder of the cap and the inner facings of the body along it's upright tube (i.e., the cap-attaching end and the dispensing port end) are configured with cooperating, tapering/conical shapes and annular sealing beads to ensure fluid flows out the dispensing port when the cap is in the “up” or open position and to prevent leakage when the cap is in the “down” or closed position.

(12) The rounded shapes of the protrusions and grooves allow the parts to be injection molded without any further machining or modification. In molding the body, a sequential ejection movement is employed so that, as the molding port/pin is pulled, the portion of the body having the protrusions will also be pulled down as the body is pushed off of the mold, thereby allowing the inner wall and outer walls to deflect over the beads and shaped grooves. Also, by forming the grooves within the cap and the corresponding protrusions directly on the body, the design is limited to only two pieces for molding, thereby simplifying molding and subsequent assembly of the tap. Thus, a method of making the dispenser tap described above is also contemplated.

(13) Specific reference is made to the appended claims, drawings, and description, all of which disclose elements of the invention. While specific embodiments are identified, it will be understood that elements from one described aspect may be combined with those from a separately identified aspect. In the same manner, a person of ordinary skill will have the requisite understanding of common processes, components, and methods, and this description is intended to encompass and disclose such common aspects even if they are not expressly identified herein.

Description

DESCRIPTION OF THE DRAWINGS

(1) Operation of the invention may be better understood by reference to the detailed description taken in connection with the following illustrations. These appended drawings form part of this specification, and any information on/in the drawings is both literally encompassed (i.e., the actual stated values) and relatively encompassed (e.g., ratios for respective dimensions of parts). In the same manner, the relative positioning and relationship of the components as shown in these drawings, as well as their function, shape, dimensions, and appearance, may all further inform certain aspects of the invention as if fully rewritten herein. Unless otherwise stated, all dimensions in the drawings are with reference to inches, and any printed information on/in the drawings form part of this written disclosure. Notably, the number and relative positioning of specific components in the Drawings (e.g., three sealing beads in FIG. **6**, two inner and outer facing camming protrusions in FIG. **2**E, the positioning of the dispensing port and tapering cone leading to the dispensing port beneath the fluid flow passage in FIG. **3**E) should all be treated as part of this written description.

(2) FIG. **1** is a three dimensional, elevated, perspective view of a dispensing apparatus according to various aspects disclosed herein.

(3) FIG. **2**A is a cross sectional, side, perspective view taken along line A-A in FIG. **1**, while FIG. **2**B is a cross sectional, front, perspective view taken along line B-B in FIG. **1**.

(4) FIG. **3**A is a side view of the valve body component in isolation as shown in FIG. **1**, while FIG. **3**B is a similar cross sectional perspective view, with the cross section taken along the central diameter of the upright tube of that same valve component. FIG. **3**C is a front view of the same valve body component in isolation, while FIG. **3**D is a cross sectional perspective view taken along the central diameter of the upright tube. FIG. **3**E is an enlarged, sectional view of FIG. **3**D highlighting the surface features imparted on the inner and outer facings thereof.

(5) FIG. **4**A is a perspective side view of the valve cap or stem component isolation as shown in FIG. **1**, and FIG. **4**B is an opposing/complimentary perspective bottom view of that same component highlighting features along the inner-most surface thereof. FIG. **4**C is a cross sectional view perspective taken along a horizontal plane defined by line C-C in FIG. **4**E (i.e., immediately beneath the top panel and adjacent to the camming threads found on the valve cap component shown in FIG. **4**A). FIG. **4**D is a top plan view and FIG. **4**E is a cross sectional perspective view taken along the central diameter of the valve cap component shown in FIG. **4**A.

(6) FIG. **5**A is an angled and partial cross sectional view of the cap valve cap of FIG. **4**A in isolation, highlighting the camming thread on the inner facing of the skirt, with the view generally taken along the plane defined by line F-F of the three dimensional inset. FIG. **5**B is an angled and partial cross sectional view of the cap valve cap of FIG. **4**A in isolation, highlighting the camming thread on the outer facing of the skirt and the comparative axial positioning of the sealing beads provided on the body (as shown in FIG. **3**E), with the view generally taken along the plane defined by line G-G of the three dimensional inset.

(7) FIG. **6** is an exploded, cross sectional, side, perspective view, drawn to scale and taken along line A-A in FIG. **1**, with separate iterations indicating the upper-most (**300**U) and lower-most (**300**L) positioning of the cap or knob relative to the body. This view also illustrates an alternative positioning for the surface features in comparison to those shown in FIG. **3**E.

DETAILED DESCRIPTION OF EMBODIMENTS

(8) As used herein, the words “example” and “exemplary” mean an instance, or illustration. The words “example” or “exemplary” do not indicate a key or preferred aspect or embodiment. The word “or” is intended to be inclusive rather an exclusive, unless context suggests otherwise. As an example, the phrase “A employs B or C,” includes any inclusive permutation (e.g., A employs B; A employs C; or A employs both B and C). As another matter, the articles “a” and “an” are generally intended to mean “one or more” unless context suggest otherwise.

(9) Food service dispensing taps is a well established and highly competitive market segment. These products are used in restaurants, caterings services, and special events, all of which prioritize low cost, simplicity of use, and (increasingly of late) sustainability. Thus, suppliers have devoted considerable effort toward developing unique products, as evidenced by the various patents noted in the Background section above.
(10) Upright, T-shaped dispensing taps with cam-actuated valves are of particular interest. These taps are preferred owing to their familiarity and ease of use, particularly in comparison to I-shaped and/or conventional screw threaded taps and valves. One of the most notable distinguishing features of T-shaped taps is the T-shaped, hollow tubular body where the horizontally disposed end attaches to the liquid container (usually a flexible bag or pouch), and the remaining end is an upright hollow member. The dispensing port is positioned at the lower/bottom end of that upright member so that fluid flows through the horizontal section and out the dispensing port, while a valve control is provided on the top end.
(11) Turning now to FIGS. **1** through **5**B, an improved, two piece, T-shaped dispensing tap **100** is disclosed. Tap **100** is formed of only two independent components: body **200** and cap or knob **300**. The significance of having two independent components is that the tap **100** may first be easily welded or attached to a flexible liner, pouch, bag, or container, after which the cap **300** may be snapped onto the body **200**.
(12) Body **200** is formed from a hollow cylindrical tubular material, preferably injection molded to allow for high volume production at the lowest possible cost. Other similar methods of manufacture could also be employed. As discussed throughout, lines A-A and B-B shown in FIG. **1** define and lie within the “horizontal plane” of the tap **100**, with “vertical” orientations falling within any plane that lies orthogonal to that horizontal plane. Notably, this horizontal plane is also orthogonally aligned relative to the central axis defined by upright tubular section **230**.
(13) Body **200** has an attachment flange **210** at one end. Flange **210** provides a flattened base **211** for coupling, while one or more series of steps **212** and/or funnels **213** define a narrowing fluid flow passage **214**. Additional flanges or formations **215** may be provided proximate flange **210** to further facilitate coupling the tap **100** to whatever appropriate liquid-carrying vessel may be required. Throughout these formations **210**, **211**, **212**, **213**, **215** the thickness of the wall may be constant or varied, depending upon the nature of the materials and manufacturing methods employed. Also, these sections will be tubular (i.e., a hollow cylinder) and, in some cases, sloped or conical, with a cross section taken in the vertical plane. While the base **211** is shown as having a circular shape owing to advantages in the process of coupling such shapes to liners, etc., other shapes (oval, curved, polygonal, curvilinear, etc.) may be used in other aspects.
(14) Notably, the length of the outer-most diameter **210***a* of flange **210** should exceed the length of any other component or part of tap **100**. Stated differently, diameter **210***a* is greater than the length or width of both body **200** and cap **300**. In one aspect, diameter **210***a* is greater than the length and width of both body **200** and cap **300** when these components are assembled (irrespective of whether cap is in the up/open position or the down/closed position).
(15) The inner surfaces of the tubular section comprising flange **210** connect to horizontal member **220**. Horizontal member **220** defines the flow channel **224**, which is connected to flow passage **214**. As shown, member **220** is a polygonal shaped tube, although other shapes are possible. Preferably, the flow capacity (i.e., the volume of liquid capable passing therethrough over a predetermined period of time) within member **220**/channel **224** is equal to or less than the flow capacity of the flange **210**/passage **214**. As with flange **210**, the thickness of the sidewalls defining member **220** may be the same or different, both in comparison to one another and/or along the length of the channel **224**.
(16) Upright tubular member **230** includes an inlet **231** to fluidically connect flow channel **224** to dispensing channel **234**. Inlet **231** is positioned axially above the dispensing port **232** but beneath the top opening **233**. Generally speaking, and subject to the further features described below,

member **230** is tapered so that at least the inner diameter, and preferably the outer diameter, of the port **232** is less than the inner diameter of the top opening **233**. Excepting for beads **250**, **260** and frusto-conical ledge **240**, this taper is regular and linear in nature (i.e., does not entail any significant curved sections. The wall thickness of member **230** may be comparatively smaller beneath the inlet **231** relative to its thickness above the inlet **230**. In one aspect, the thickness of the wall in member **230** above the inlet **231** is substantially constant so as to allow for reliable operation of camming protrusions **270** with camming surfaces **370**.

(17) Bevel or cone **234** may be formed at the inner facing of the top opening **233** to facilitate coupling and sealing engagement with the cap **300**. Immediately beneath but still axially offset beneath the point where **234** meets the inner facing of the member **230**, camming protrusions **270** extend radially inward and outward from the top wall section **235**.

(18) Preferably, there are a plurality of protrusions **270** spaced apart at regular intervals so as to align and cooperate with the surfaces **370** provided on cap **300**. As shown and in a preferred embodiment, two sets of protrusions **270***a*, **270***b* are provided. The features indicated below for a single protrusion **270** are applicable to any protrusions that may be provided. Also, it is important to note that no other protruding features can be provided on either facing of wall **235**, as any additional features may interfere with the smooth and intended operation of the valve mechanism.

(19) Each protrusion **270** includes an inner facing **271** and an outer facing **272**. The radial extension or reach of facing **271** is shorter than that of facing **272**. Further, the shape of facing **271** is preferably of a rounded button or bump, whereas facing **272** presents as a more defined, raised cylinder. The vertical facings **271***a*, **272***a* should be spherical or rounded in nature so as to cooperate with and move smoothly along and within surface **370**. Further, these rounded facings **271***a*, **272***a* facilitate ejection of the body **200** from its manufacturing mold, as well as allow for easier assembly of the cap **300** onto and into the upright member **230**.

(20) Axial offset beneath protrusions **270** but above the inlet **231**, a series of sealing beads **250**, **260** are provided. Each of beads **250**, **260** form separate, complete raised annuls along the inner facing of the member **230**. In a preferred arrangement, beads **250**, **260** are circularly aligned within separate horizontal planes. The inner diameter of lower bead **250** is smaller than that of the upper bead **260**. Further, a tangent T can be drawn between the top edges of beads **250**, **260** in a manner that matches the taper of the inner surfaces of the member **230** from the top opening **233** all the was to the beginning of the ledge **240**.

(21) Particularly with respect to lower bead **250**, the purpose of beads **250**, **260** is to sealingly but slidingly engage the surface of the cylindrical member **330** of cap **300**. Thus, each bead **250**, **260** will be appreciably raised and distinguishable from the smooth, tapered sidewall on either side of it; however, its profile will still be conducive to allowing the member **330** to glide across/over it. In this manner, fluid flowing along the path established by **214**, **224**, **234** will not inadvertently leak out of the top opening **233**.

(22) Another aspect of the sealing beads **250**, **260** can be discerned by comparing FIGS. **3**E and **6**. In the former, upper sealing bead **260** is positioned to retain a sealing surface against the knob **300**, whereas a plurality (two or more) axially offset lower sealing beads **250** are provided in the latter. In FIG. **6**, the upper sealing bead **260** is also repositioned axially so that its sealing circumference remains beneath the position of the grooves **370**, thereby insuring that the body **200** and cap **300** remain sealed at all times during the twisting the cap **300** relative to the body **200** (i.e., actuation to open and close the valve).

(23) Immediately beneath inlet **231**, the profile of the inner surface of member **230** constricts appreciably. That is, a ledge or differently sloped, conical section **240** serves as an axial stop that cooperates with section **340** on the member **330**. The bottom-most edge **241** of the section **240** also sealingly engages ramp **341** when the cap **300** is the closed/down position so as to prevent unwanted leakage or loss of fluid through the tap **100** and, more specifically, by way of path **214**, **224**, **234** and out port **232**. Thus, the inner diameter of the tube **230** within section **240** gradually

reduces at a steeper slope/angle than the normal taper of that tube **230**. At its narrowest point (i.e., proximate the sealing surface **241**), the inner diameter of section **240** is smaller than the corresponding inner diameters of either bead **250**, **260**.

(24) The lower wall section **242** runs from the bottom edge of the cone section **240** to the terminal edge at port **232**. The wall thickness in this section is preferably reduced as it approaches the edge at port **232** (i.e., it is thicker closer to the inlet **231** and thinner at the port **232**). Section **242** is also tapered, again preferably at an angle that is different and steeper than the taper along section **235** but less than that of the cone section **240**.

(25) As noted above, the overall purpose of the ever-narrowing inner diameter along upright member **230** (i.e. from top opening **233** to port **232**) is to ensure member **230** engages cap **300** and, more specifically, cylinder **330**. To that end, cylinder **330** should have sufficient pliability and resilience to maintain a sealing fit with at least one of beads **250**, **260** at all times (i.e., whether in the open or closed position). The increased taper in section **242** could also serve to form a plug seal with the lower facings **342** in section **340**.

(26) Cap **300** is formed as a discrete element in comparison to body **200**. Cap **300** presents as an outer-most annular skirt **310** with a coaxially aligned sealing cylinder **330** set within and extending down axially further than the bottom edge **311** of skirt. Skirt **310** and cylinder **330** are connected by a panel **360**, with at least the outer most periphery **361** of panel **360** connecting the top edge **312** of the skirt **310** to the top end **332** of the cylinder **330**. Notably, the sidewall **313** of skirt **310** and sidewall **333** of the cylinder **330** are in a substantially parallel arrangement to define a radial gap **320**.

(27) Gap **320** is sized and configured to receive the thickness of wall section **235**, while also allowing the cap **300** to rotate freely about the upright member **230**. Specifically, at least along the sections where caroming surfaces **370** are provided, gap **320** should define a substantially constant and/or non-varying distance between skirt **310** and cylinder **330**. Proximate to end **311**, the gap **320** may flare outward so as to increase in distance, as this should facilitate coupling the cap **300** to the upright member **230**.

(28) Panel **360** is formed primarily in the horizontal plane, although it may be imparted with a raised or round appearance. Also, panel **360** may present as a substantial flat outer surface so as to enable forming indicia relating to operation, contents, and/or the source of origin for the tap **100** and/or the fluid being dispensed thereby. Notably, the inner surfaces of cylinder **330** need not be joined at the top end **332** and, either alternatively or additionally, these sidewalls **333** could be joined at lower position. It may also be possible to form panel **360** as an annular ring so that this hollowed inner portion of the cylinder **330** is accessible, possibly for cleaning or aesthetic purposes.

(29) When top panel **360** continuously encloses the inner portion of cylinder **330**, a plurality of support ribs **362** may be formed along the inner surface of wall **333**. Ribs **362** provide sufficient strength to ensure cap **300** can withstand rotational forces and, more generally, serve its intended purpose as a sealing element.

(30) Conversely, the lower portion **340** of cylinder **330** includes a ramp section **341** and, in comparison to wall **333**, a thinner, more flexible, and resilient wall section **342**. The outer diameter of section **342** may taper inward (i.e., decrease) so as to cooperate with the shape/taper of section **242**. More generally, the lower extremities of cylinder **330**, including but not limited to section **342**, can temporarily flex or constrict inward in response to the sealing faces and/or beads formed on the inner surfaces of member **230**.

(31) As noted above, camming surfaces **370** are formed along arcuate sections of opposing surfaces on skirt **310** and cylinder **330**. In particular, camming surface **317** is formed on an inner facing of skirt **310**, while camming surface **337** is formed on an outer facing of cylinder **330**. In both instances, surfaces **317**, **337** are proximate one another so as to receive and cooperate with protrusions **270** noted above.

(32) Surfaces **370** are formed as spiral grooves **318**, **338**. The pitch of these grooves are similar and sufficient, when considered in combination with their length, to allow the cap **300** to rotatably travel in an axial distance that blocks or opens inlet **231**. Thus, in the down position, sealing faces **241**, **341** are engaged so as to close off inlet **231** and prevent liquid from being drawn (via gravity) from path section **224** into channel **234**. In the up position, inlet **231** is unobstructed so that liquid may flow freely through channel **234** and out of dispensing port **233**.
(33) The depth and shape/profile of grooves **317**, **337** will cooperate with the radial extension of the protrusions **270** so as to define a traveling path as the cap **300** is rotated relative to body **200**. At the open and/or closed positions (which roughly correspond to points at or near the top and bottom edges of the grooves **317**, **337**), a seat **319** can be formed as a further indent. The shape of seat **319** matches the vertical presentation of the protrusion **270** so as to provide a natural fit and resting place, and such seats **319** may be at the upper and/or lower positions of the grooves **370**. In operation, this provides the user with a tactile sensation of where the open and closed positions reside within tap **100**. The use of opposing sets of protrusion-and-groove combinations also ensures that rotation of the cap has a smooth sensation, with the seats providing a tactile, and possibly even audible, sensation so that the user knows when the open position has been reached. Further, by relying upon circular protrusions **270** and corresponding grooves **370** in combination with a plurality of opposing camming surfaces **317**, **337**, the cap **300** can be guided along its spiraling path easily and without risk of decoupling.
(34) The pitch of the grooves **317**, **337** can be measured relative to the circumference of the circular surface on which they reside. To that end, grooves **317**, **337** travel radially less than one full turn (i.e., less than 360° around that circumference) and, more preferably, their arc is great than about 45° but less than 180° (and, still even more preferably, between 75° and 135°, with about 90° for two grooves disposed on opposing facings of the skirt and sealing cylinder being most preferred). Relative to the axial height, the grooves **317**, **337** will encompass an axial height that is less than the corresponding flat surfaces of the sidewall **313** of skirt **310** (i.e., from above end **311** to top **312**) and the sidewall **333** of cylinder **330** (i.e. from above ramp section **341** to top **332**) on which they are formed, with the grooves covering at least 50%, between 60% and 90%, and most preferably about two thirds to three quarters of these respective surfaces. The grooves and protrusions may have cooperating spherical profiles to facilitate their sliding movement and seating/coupling.
(35) The number and location of protrusions **270** should match the number of camming surfaces **370**. While a pair of protrusions and surfaces are illustrated and believed to represent the most desirable balance of operational and manufacturing ease, it will be understood that one or a plurality of these features may be provided. Also, while protrusions **270***a*, **270***b* are shown to be at the same axial elevation, they do not necessarily need to be provided as such. If less than two protrusions are used, it may denigrate the preferred smooth motion and tactile sensation of seating the protrusion, while three or more protrusions may limit the variety of injection molding materials that can be used and require a more aggressive geometry (in terms of pitch, positioning, etc.) to produce the desired characteristics for operating (i.e., opening and closing) the tap.
(36) On the outer surface of skirt **310**, ribs, knurling, and/or flattened faces may be formed. These formations will enhance a user's ability to grip and twist the cap, while also providing an indication of how far the cap has been (or needs to be) rotated.
(37) The structures described above are comparatively easy to mold, manufacture, and assemble, particularly in comparison to the more complex designs noted in the Background section above. The shape and positioning of the protrusions **270** are particularly amenable to sequential ejection from a molding dye. Also, the configuration and coupling of cap **300** to body **200** should allow for natural fit and alignment.
(38) Separately, the final, assemble tap **100** does not include any open grooves or other areas where dirt or mold can accumulate as the tap **100** is used. Further, by concealing the camming action, it

reduces the possibility of having an obstruction impede rotation and operation of the cap **300**. Thus, the skirt **310** of the cap **300** presents as a continuous solid surface extending down from the top panel **360**, thereby completely concealing the camming protrusions **270** and camming grooves **370** irrespective of their rotational/axial positioning (i.e., at an upper, open configuration or a lower, closed configuration).

(39) In addition to the foregoing description, aspects of the invention may include a dispensing tap having any combination of the following features: a hollow tubular body having a T-shape with a narrowing funnel portion defining an container attachment end, a tapering sidewall defining a dispensing port end, an upper sidewall defining a cap-receiving end, and a flow channel fluidically connecting the container attachment end to the cap-receiving end and the dispensing port end and wherein: (i) an outer facing of the upper sidewall is provided with at least one radially extending camming protrusion, (ii) an inner facing of the upper sidewall is provided with at least one annular sealing bead and at least one radially extending camming protrusion, and (iii) an inner facing of the tapering sidewall is provided with an annular sealing bead; a cap, fitted over the cap-receiving end, having an skirt coaxially fitted around an sealing cylinder and a top panel connected to top terminal edges of the skirt and the sealing cylinder and wherein: (a) at least one camming groove spirals axially along each of an inner facing of the skirt and an outer facing of the sealing cylinder, (b) a bottom terminal edge of the sealing cylinder extends axially beyond a bottom terminal edge of the skirt, and (c) the upper sidewall of the body is received in a gap formed between the skirt and the sealing cylinder; wherein at least one of the radially extending camming protrusions has a circular, spherical shape; wherein all of the camming protrusions provided on outer facing of the upper sidewall are larger than the camming protrusion(s) provided on the inner facing of the upper sidewall; wherein a plurality of annular sealing beads are axially offset along the inner facing of the upper sidewall; wherein the annular sealing bead of the upper sidewall is positioned to remain axially beneath the camming grooves irrespective of the positioning of the camming protrusions within the camming grooves; wherein at least one of the camming grooves is provided with a seat; wherein the sealing cylinder includes a tapered lower section that conforms to the tapering sidewall when the cap is rotated to a closed position; wherein a conical section constricts radially inward beneath the annular sealing bead provided on the tapering sidewall; wherein the camming protrusion on the outer facing is formed adjacent and opposite to the camming protrusion on the inner facing; wherein two camming protrusions are equally spaced apart along a circumference on the inner and outer facings of the upper sidewall; wherein all of the camming grooves are configured to position the cap in an open or closed position at less than one full rotation of the cap relative to the body; wherein a radial attachment flange is provided at the container attachment end; wherein a sealing bevel is provided at a terminal end of the upper sidewall; and wherein the camming protrusions and the camming grooves are concealed in the gap of the cap.

(40) References to coupling, connection, or attachment in this disclosure are to be understood as encompassing any of the conventional means used in this field. This may take the form of snap- or force fitting of components having tabs, grooves, and the like. Nevertheless, threaded connections, annular or partial bead-and-groove arrangements, cooperating cam members, and slot-and-flange assemblies could be employed. Adhesive and fasteners could also be used, although such components must be judiciously selected so as to retain the recyclable nature of the assembly.

(41) In the same manner, engagement may involve coupling or an abutting relationship. These terms, as well as any implicit or explicit reference to coupling, will should be considered in the context in which it is used, and any perceived ambiguity can potentially be resolved by referring to the drawings.

(42) All components should be made of materials having sufficient flexibility and structural integrity, as well as a chemically inert nature. The materials should also be selected for workability, cost, and weight. Common polymers amenable to injection molding, extrusion, or other common forming processes are useful, although a single grade is preferred. As such, polypropylene is

expected to have particular utility.
(43) In fact, another reason consumers, manufacturers, and others will find utility in these designs/components is precisely because of the use of only a single grade of polymer (e.g., polypropylene). This approach should simplify both manufacturing and recycling of the dispenser apparatus. Other materials—and particularly recyclable, injection molding materials—may be useful, including without limitation polyethylene (including low density and other grades), polystyrene (including high impact and other grades), acrylonitrile butadiene styrene, and polyacetals (including polyoxymethylene, polyacetal, polyformaldehyde, and other grades).
(44) Although the present embodiments have been illustrated in the accompanying drawings and described in the foregoing detailed description, it is to be understood that the invention is not to be limited to just the embodiments disclosed, and numerous rearrangements, modifications and substitutions are also contemplated. The exemplary embodiment has been described with reference to the preferred embodiments, but further modifications and alterations encompass the preceding detailed description. These modifications and alterations also fall within the scope of the appended claims or the equivalents thereof.

Claims

1. A dispensing tap comprising: a hollow tubular body having a T-shape with a narrowing funnel portion defining a container attachment end, a tapering sidewall defining a dispensing port end, an upper sidewall defining a cap-receiving end, and a flow channel fluidically connecting the container attachment end to the cap-receiving end and the dispensing port end and wherein: (i) an outer facing of the upper sidewall is provided with at least one radially extending outer camming protrusion, (ii) an inner facing of the upper sidewall is provided with at least one annular sealing bead and at least one radially extending inner camming protrusion, and (iii) an inner facing of the tapering sidewall is provided with an annular outlet sealing bead, wherein the outer camming protrusion on the outer facing is formed adjacent and opposite to the inner camming protrusion on the inner facing; and a cap having an skirt coaxially fitted around an sealing cylinder and a top panel connected to top terminal edges of the skirt and the sealing cylinder and wherein: (a) at least one camming groove spirals axially along each of an inner facing of the skirt and an outer facing of the sealing cylinder, (b) a bottom terminal edge of the sealing cylinder extends axially beyond a bottom terminal edge of the skirt, and (c) when the cap is fitted over the upper sidewall of the body, the cap-receiving end of the upper sidewall is positioned in a gap formed between the skirt and the sealing cylinder, and the inner and outer camming protrusions are received in the at least one camming groove.
2. The dispensing tap of claim 1 wherein all of the radially extending camming protrusions has a spherical shape conforming to a shape of each camming groove associated therewith.
3. The dispensing tap of claim 1 wherein each of the outer camming protrusions provided on outer facing of the upper sidewall are larger than the inner camming protrusion(s) provided on the inner facing of the upper sidewall.
4. The dispensing tap of claim 1 wherein a plurality of annular sealing beads are axially offset along the inner facing of the upper sidewall.
5. The dispensing tap of claim 1 wherein the annular sealing bead(s) of the upper sidewall is positioned to remain axially beneath the camming grooves irrespective of the positioning of the camming protrusions within the camming grooves.
6. The dispensing tap of claim 1 wherein at least one of the camming grooves is provided with a seat denoting an open and/or closed position for the dispensing tap.
7. The dispensing tap of claim 1 wherein the sealing cylinder includes a tapered lower section that conforms to the tapering sidewall when the cap is rotated to a closed position.
8. The dispensing tap of claim 1 wherein a conical section constricts radially inward beneath the

annular sealing bead on the tapering sidewall.

9. The dispensing tap of claim 1 wherein a second outer camming protrusion and a second inner camming protrusions are equally spaced apart along a circumference on the inner and outer facings of the upper sidewall.

10. The dispensing tap of claim 1 wherein all of the camming grooves are configured to position the cap in an open or closed position at less than one full rotation of the cap relative to the body.

11. The dispensing tap of claim 1 wherein a radial attachment flange is provided at the container attachment end.

12. The dispensing tap of claim 1 wherein a sealing bevel is provided at a terminal end of the upper sidewall.

13. The dispensing tap of claim 1 wherein the inner and outer camming protrusions and the at least one camming groove are concealed in the gap of the cap.
